FEA-Based Design Optimization of Rectangular Vacuum Vessel Flange for Spherical Tokamak

A Project Report submission as a part of
Summer School Program-2026

Submitted By: -

Harsh Pratap Singh

B.Tech. in Mechanical Engineering, DIT University, Dehradun

Under The Guidance Of

Mr. Vishal Verma

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Spherical Tokamak Division, IPR

INSTITUTE FOR PLASMA RESEARCH
(An Aided Institute of Department of Atomic Energy, Government of India)
Bhat, Gandhinagar – 382 428 Gujarat

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Abstract

Vacuum vessel flanges are critical components in all tokamak systems, ensuring leak-tight connections between vacuum vessels and diagnostic or pumping ports. When ConFlat (CF) ports are welded onto thin rectangular plates, distortion often occurs due to thermal stresses, bolt pretension, differential pressure loading and other dynamic load due to electromagnetic force. This project aims to optimize the thickness of a rectangular stainless steel flange (450 mm × 560 mm) to minimize distortion under welding and operational loads.

The project will provide students with hands-on experience in computational mechanics applied to tokamak component design. They will learn how welding sequence and boundary conditions influence distortion, how bolt pretension interacts with plate stiffness, and how vacuum loading affects flange integrity. By performing FEA simulations, students will gain practical skills in modeling, meshing, thermal–structural coupling, and optimization. The case study of plate bending will be explored in detail, bridging theoretical mechanics with real-world tokamak engineering challenges. This project will prepare students to contribute to advanced vacuum system design in fusion research facilities.

Scope of Work:

- Literature review on welding distortion in fusion-related vacuum vessel components.
- CAD modeling of 13 mm, 16 mm, 19 mm, and 22 mm flange thicknesses.
- Transient thermal and static structural FEA with welding sequence comparison and bolt pretension / equivalent axial clamping load.
- Optimization of thickness to identify the best flange configuration.

Acknowledgment

I would like to express my sincere gratitude to **Mr. Vishal Verma**, Project Guide, **Spherical Tokamak Division, Institute for Plasma Research (IPR), Gandhinagar**, for his continuous guidance, valuable suggestions, technical discussions, and constant encouragement throughout the Summer School Program 2026. His support and feedback played a significant role in the successful completion of this project.

I would also like to thank the **Institute for Plasma Research (IPR)** for providing me with the opportunity to participate in the Summer School Program 2026 and for exposing me to practical engineering problems related to spherical tokamak vacuum vessel design. The experience gained during this program has significantly enhanced my understanding of finite element analysis and engineering design optimization.

I am grateful to **DIT University, Dehradun**, especially the Department of Mechanical Engineering, for providing me with a strong academic foundation that enabled me to successfully undertake this project.

Finally, I would like to thank all the scientists, engineers, technical staff, faculty members, and fellow interns at IPR for their support, valuable interactions, and encouraging learning environment throughout the program. The knowledge and experience gained during this project will remain an important milestone in my academic and professional journey.

1.Introduction:

Rectangular vacuum vessel flanges are critical sealing and structural components in spherical tokamak systems, where they provide leak-tight connections between the vacuum vessel and diagnostic or pumping ports while maintaining the required vacuum integrity during operation. Since these flanges are subjected to welding during fabrication and multiple mechanical loads during service, their structural integrity and dimensional stability are of great importance. Even a small amount of welding-induced distortion may affect flange flatness, sealing performance, and the overall reliability of the vacuum system.

The rectangular stainless steel flange considered in this study consists of ConFlat (CF) ports welded onto a base plate. During the welding process, These thermal effects are commonly referred to as welding-induced distortion and residual stresses, which directly influence the sealing performance and structural integrity of the flange. Localized heating and subsequent cooling generate non-uniform temperature distribution, resulting in thermal distortion and residual stresses. In addition to these welding effects, the flange experiences bolt pretension, represented in this work as an equivalent axial clamping load**,** together with atmospheric pressure loading acting under vacuum operating conditions. The combined influence of these thermo-mechanical loads significantly affects the deformation, stress distribution, and overall structural response of the flange.

Experimental evaluation of several flange thicknesses and welding sequences is both time-consuming and expensive. Therefore, Finite Element Analysis (FEA) provides an efficient and reliable approach for predicting the thermo-structural behavior of the flange before manufacturing. By combining transient thermal analysis with static structural analysis, different design configurations can be evaluated under identical loading conditions, enabling systematic comparison and optimization.

In the present work, parametric CAD models of rectangular vacuum vessel flanges having **13 mm, 16 mm, 19 mm, and 22 mm** plate thicknesses were developed. Three different welding sequences, namely **Sequence A (Anti-Clockwise), Sequence B (Symmetric/Opposite Side), and Sequence C (Alternate Cross-Pattern)**, were investigated to evaluate their influence on the thermo-mechanical response of the flange.. The thermal history obtained from the transient thermal analysis was subsequently imported into the static structural analysis, where equivalent axial clamping load (representing bolt pretension), atmospheric pressure, and appropriate support conditions were applied.

The primary objective of this study was to identify the optimum flange configuration by minimizing total deformation while maintaining acceptable equivalent von-Mises stress and equivalent strain. The final design selection was based on a comprehensive comparison of different plate thicknesses and welding sequences under identical thermo-mechanical loading conditions, providing a practical design recommendation for spherical tokamak vacuum vessel flange applications..

2.Theoretical/ Simulation/ Experimental Details:

2.1. CAD Geometry:

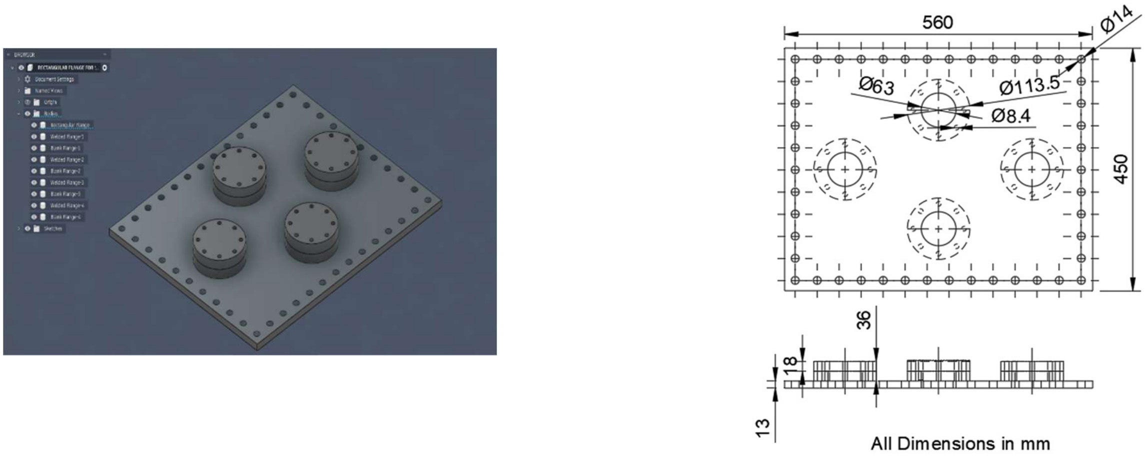


Figure 1. 13 mm Rectangular Vacuum Vessel Flange with CF-63 Flanges

The CAD model of the **13 mm thick** rectangular vacuum vessel flange was developed in Autodesk Fusion 360 as the minimum plate thickness considered in the present investigation. The overall geometry, including the four CF-63 ports, bolt-hole arrangement, and flange dimensions, was kept identical to the remaining models. This configuration was selected to evaluate the influence of lower plate stiffness on thermal distortion and structural response during thermo-mechanical loading. The model was subsequently imported into ANSYS Mechanical for finite element analysis.

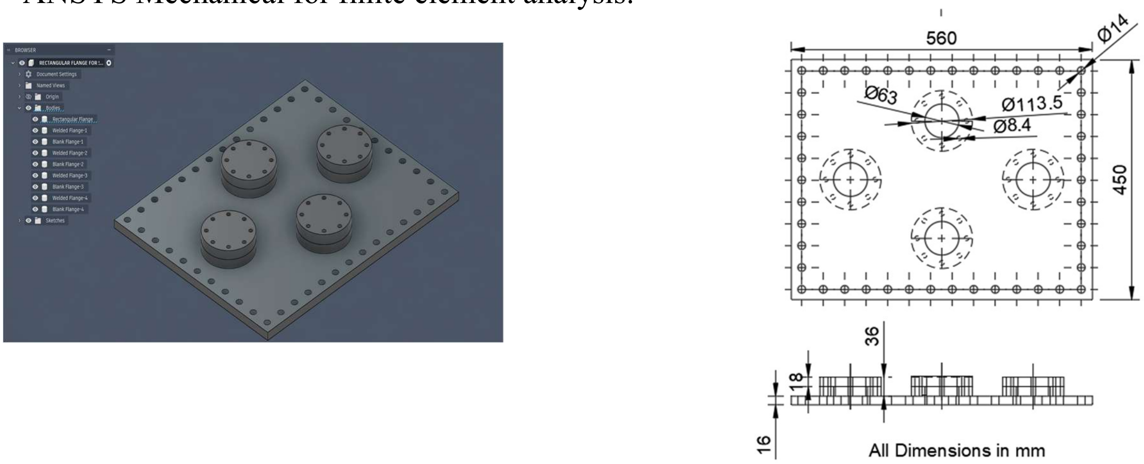


Figure 2. 16 mm Rectangular Vacuum Vessel Flange with CF-63 Flanges

The **16 mm thick** flange model was created using the same geometric configuration while increasing only the plate thickness. This configuration was investigated to improve structural rigidity without introducing unnecessary weight. Later simulation results demonstrated that this thickness, combined with the **Sequence B (Symmetric/Opposite Side)** welding sequence, provided the most balanced thermo-mechanical response among all investigated configurations.

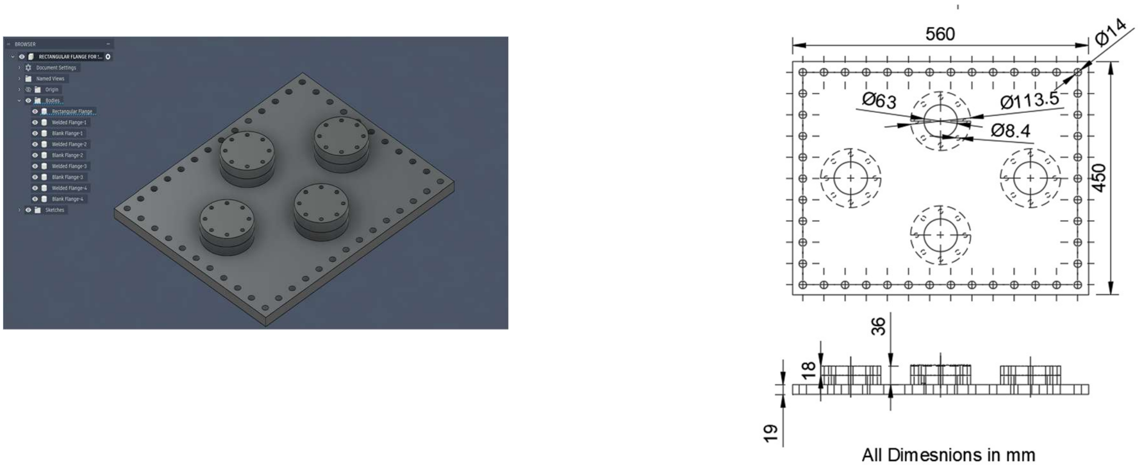


Figure 3. 19 mm Rectangular Vacuum Vessel Flange with CF-63 Flanges

The CAD model with a **19 mm plate thickness** was developed to investigate the effect of further increasing structural stiffness. Except for the plate thickness, all geometric features remained unchanged to ensure a fair comparison under identical loading and boundary conditions. The model was analyzed to evaluate whether the additional thickness could further reduce deformation while maintaining acceptable stress and strain levels.

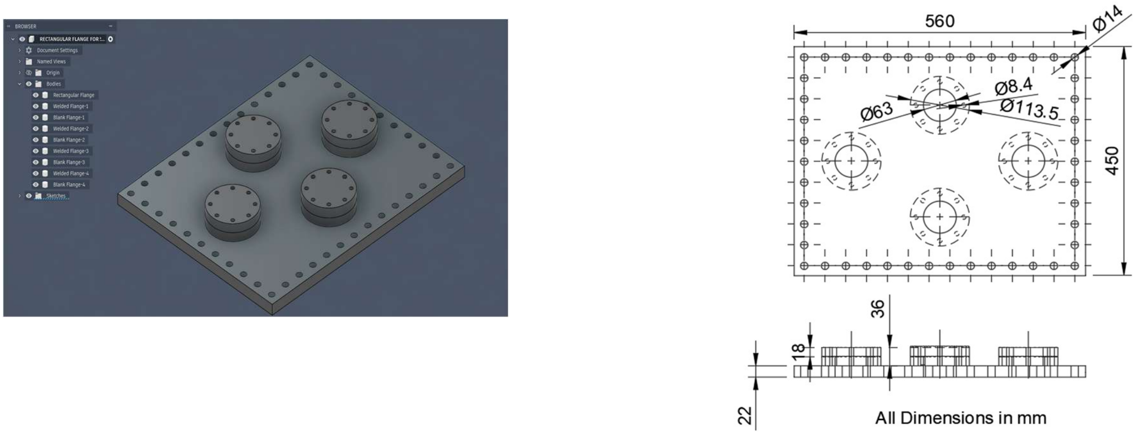
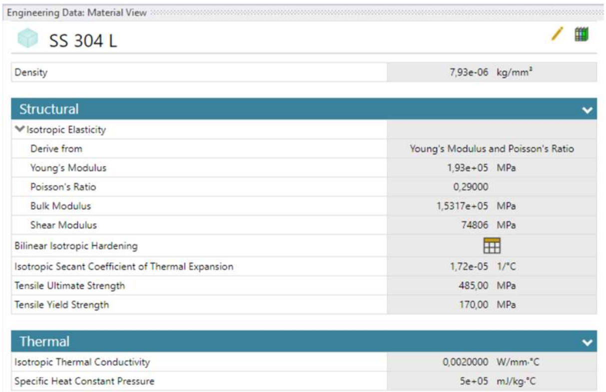


Figure 4. 22 mm Rectangular Vacuum Vessel Flange with CF-63 Flanges

The **22 mm thick** flange represented the maximum plate thickness considered in the parametric study. This model was included to examine the structural behavior of a comparatively thicker flange under identical thermo-mechanical loading conditions. Although the increased thickness improved stiffness, its overall structural response was compared with the other configurations to determine whether additional material offered any practical advantage in the final optimized design.

2.2. Material Assignment & Mesh:

SS304L stainless steel was assigned as the material model for the rectangular vacuum vessel flange, and the required thermal and mechanical properties were imported from the ANSYS Engineering Data library. The material definition was kept consistent for all flange thicknesses so that the structural response could be compared under identical thermo-mechanical loading conditions.


Engineering Data: Material View

SS 304 L

Density	7,93e-06 kg/mm³
Structural	
Isotropic Elasticity	
Derive from	Young's Modulus and Poisson's Ratio
Young's Modulus	1,93e+05 MPa
Poisson's Ratio	0,29000
Bulk Modulus	1,5317e+05 MPa
Shear Modulus	74806 MPa
Bilinear Isotropic Hardening	
Isotropic Secant Coefficient of Thermal Expansion	1,72e-05 1/°C
Tensile Ultimate Strength	485,00 MPa
Tensile Yield Strength	170,00 MPa
Thermal	
Isotropic Thermal Conductivity	0,0020000 W/mm·°C
Specific Heat Constant Pressure	5e+05 mJ/kg·°C

Figure 5. Material assignment of SS304L stainless steel for the rectangular vacuum vessel flange model in ANSYS Mechanical.

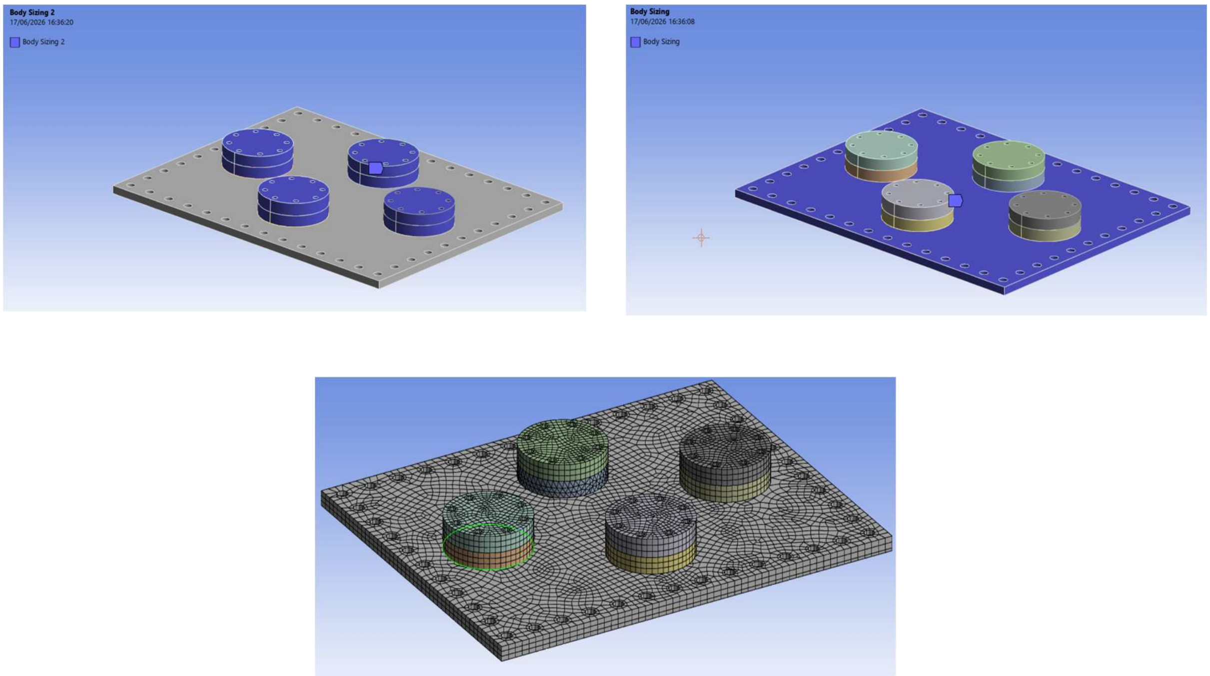


Figure 6. Finite element mesh generation of the rectangular vacuum vessel flange using body sizing in ANSYS Mechanical

A body-sizing approach was used to generate the finite element mesh for the flange model, with local refinement near the CF flange and weld regions to improve solution accuracy. A refined 3D solid mesh was adopted to capture the thermal gradients and structural response more effectively before performing the transient thermal and static structural analyses

2.3. Thermal Boundary Conditions & Welding Sequence:

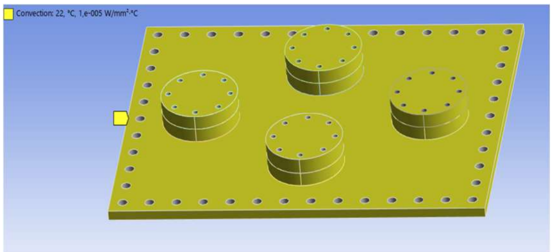


Figure 7. Convection boundary condition applied on the rectangular vacuum vessel flange during transient welding simulation.

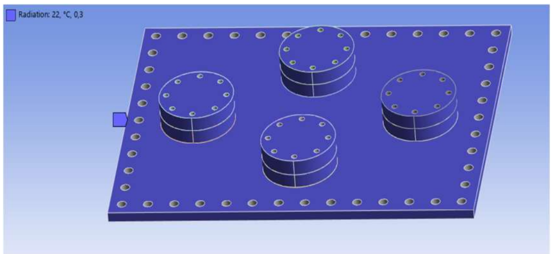


Figure 8. Radiation boundary condition applied on the rectangular vacuum vessel flange during transient welding simulation.

Convection coefficient: h = 10 W/m²°C (Natural convection, still air, horizontal flat plate — standard value for shop floor welding environment) Ambient temperature: 22°C. Convection boundary condition was applied on the exposed surfaces of the flange to represent heat loss from the weld region to the surrounding environment. This condition helps simulate the gradual removal of thermal energy during the welding process and contributes to a more realistic transient thermal response of the flange.

Emissivity: ε = 0.3 (As-received machined SS304L surface — standard reference value for unpolished austenitic stainless steel). Radiation boundary condition was applied to account for heat loss from the flange surfaces through thermal radiation during welding. Including radiation makes the thermal model more realistic by capturing an additional heat-transfer mode that acts simultaneously with convection.

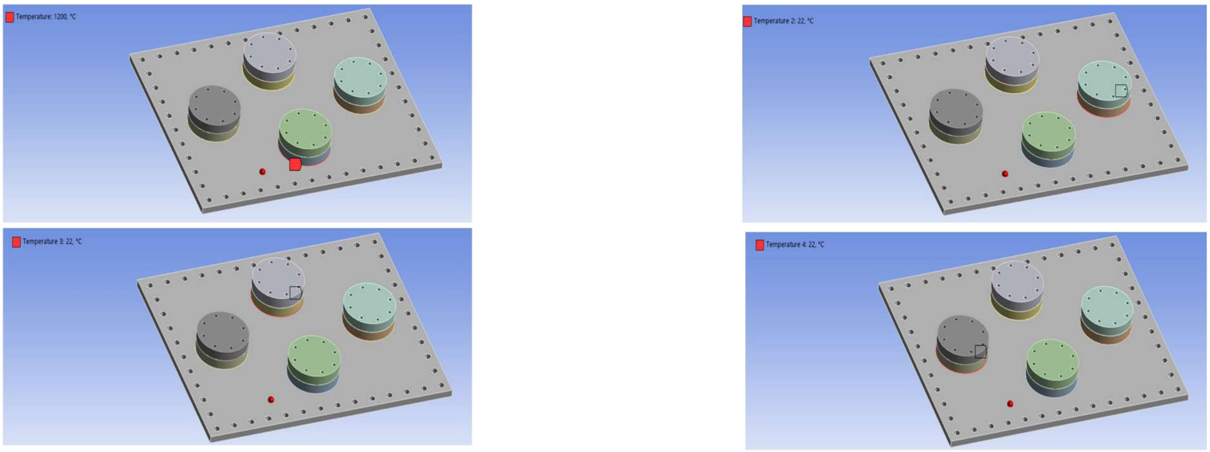


Figure 9. Sequential temperature application at the CF port locations during transient welding simulation.

The welding heat input was applied sequentially at the CF port locations to represent the transient heating stage of the welding process. This sequence-wise temperature application allowed the heat source to move through the flange in a controlled manner, creating localized heating near the weld zone and a corresponding thermal gradient in the surrounding material.

Weld perimeter per CF port:
L = π × D = π × 113.5 = 356.6 mm
Standard TIG welding speed for SS304L:
v = 120 mm/min
Time per port:
t = L / v = 356.6 / 120
= 2.97 min ≈ 180 seconds
Total simulation time:
4 ports × 180 s = 720 seconds = 4 time steps

SIMULATION SIMPLIFICATION: — In ANSYS transient thermal analysis, each welding pass was represented as: → 1 second per time step (active heat) → 4 time steps total = 4 seconds Reason for simplification: The objective of this simulation was to capture the SEQUENCE of heat application and resulting thermal gradient — not the absolute time duration. The relative order of thermal loading between ports is preserved exactly. This is a standard simplification in FEA-based welding distortion studies (consistent with Biswas et al. 2010 approach).

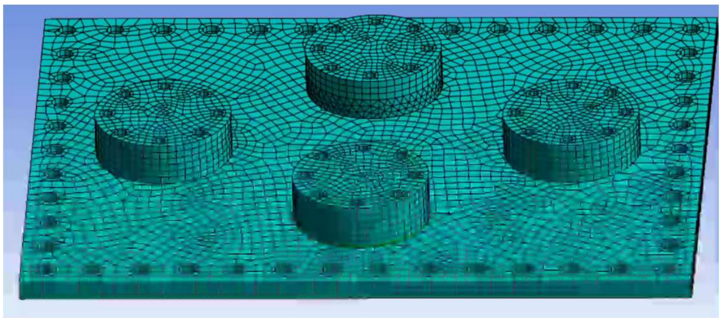


Figure 10. Welding Sequence A: Anti-Clockwise (Baseline)

Sequence A was taken as the baseline welding order, in which the CF ports were welded in an anti-clockwise sequence. This configuration was used as the reference case to study the effect of a conventional welding path on distortion and stress redistribution in the rectangular vacuum vessel flange. By keeping the plate thickness constant, the thermal response and resulting deformation pattern produced by this sequence could be directly compared with the other welding orders

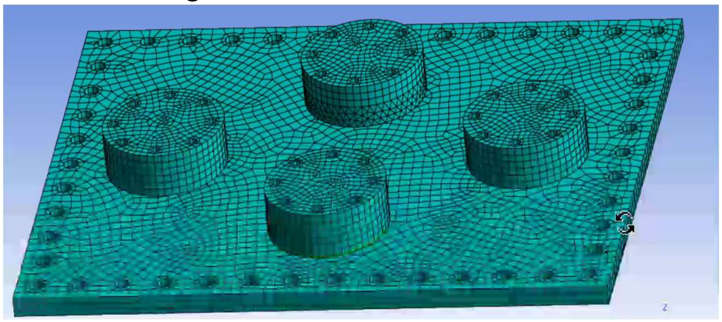


Figure 11. Welding Sequence B: Symmetric /Opposite side

Sequence A was taken as the baseline welding order, in which the CF ports were welded in an anti-clockwise sequence. This configuration was used as the reference case to study the effect of a conventional welding path on distortion and stress redistribution in the rectangular vacuum vessel flange. By keeping the plate thickness constant, the thermal response and resulting deformation pattern produced by this sequence could be directly compared with the other welding orders

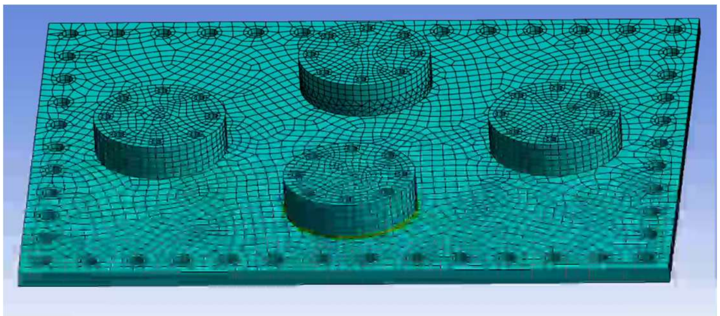


Figure 12. Welding Sequence C: Alternate Cross-Pattern

Sequence C used an alternate cross-pattern welding order, in which the weld path was changed from one side to the opposite side in a crosswise manner. This sequence was included to examine whether alternating the welding locations could further influence the distortion pattern and reduce residual stress concentration. The results from this case were compared with Sequences A and B to identify the welding order that gave the best overall thermo-mechanical performance.

2.4. Transient Thermal Analysis:

Temperature Distribution:

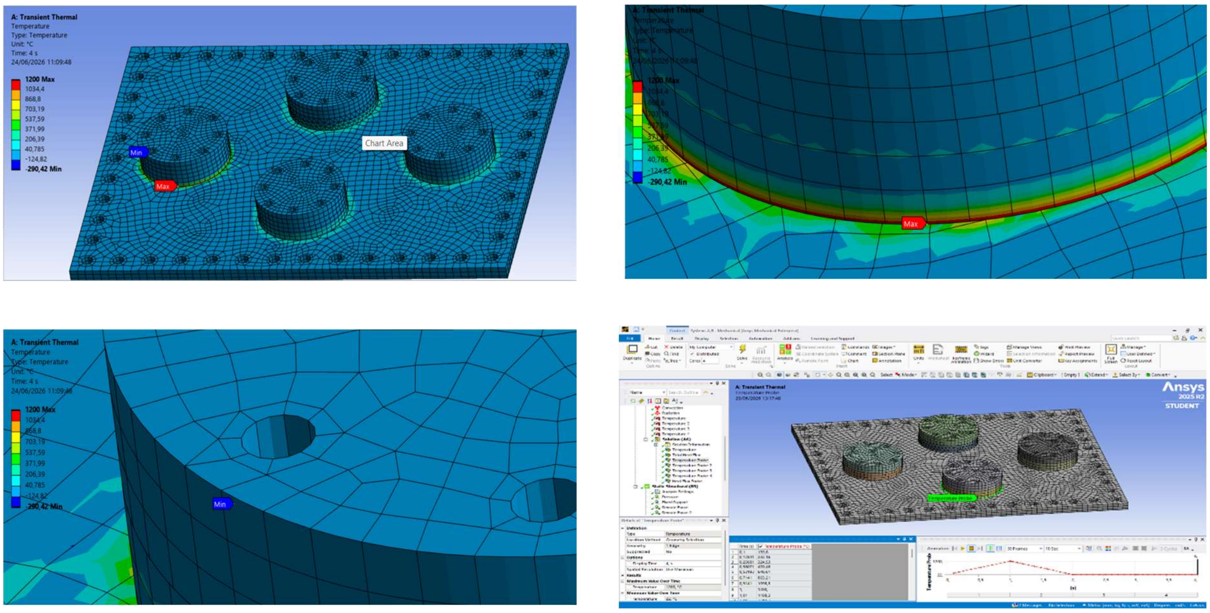


Figure 13. Temperature distribution, temperature probe history, and heat flux distribution during transient welding simulation of the optimized 16 mm flange using Sequence B

The temperature distribution obtained from the transient thermal analysis of the optimized **16 mm flange using Sequence B (Symmetric/Opposite Side)** shows the localized effect of the moving welding heat source on the rectangular vacuum vessel flange. As the heat source is applied sequentially on the CF-63 ports, a steep thermal gradient is generated in the region surrounding the active weld zone. The maximum temperature is concentrated near the weld bead, whereas the temperature decreases gradually as the distance from the heat input increases. This clearly indicates that the thermal energy is not uniformly distributed across the flange, but is instead highly localized around the welding region during the transient process. The contour pattern also shows that the plate away from the weld path remains relatively cooler, which is expected because the surrounding material acts as a heat sink and dissipates the input energy through conduction, convection, and radiation. Such non-uniform heating is one of the primary reasons for welding-induced distortion and residual stress formation in vacuum vessel flanges. A temperature probe was included at the weld location to capture the actual thermal cycle and verify the heating and cooling history during welding.

Heat Flux Distribution:

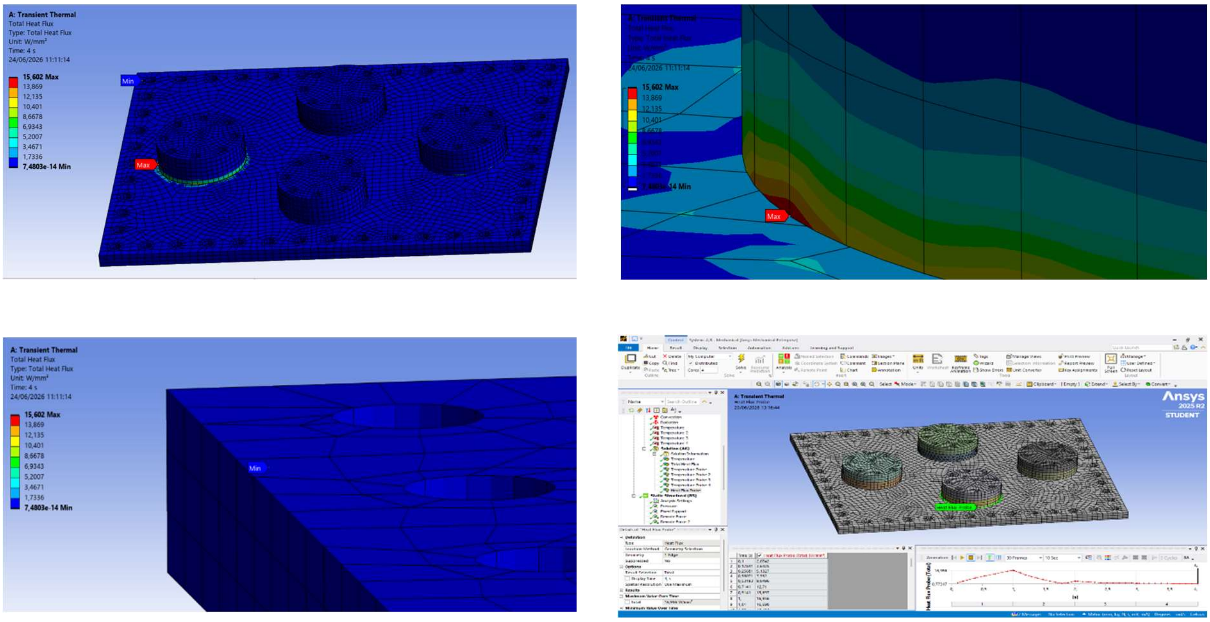


Figure 14. Heat flux distribution during transient welding simulation of the optimized 16 mm flange using Sequence B.

The heat flux distribution corresponding to the same optimized **16 mm flange with Sequence B** describes how the thermal energy flows through the flange material during welding. The maximum heat flux is observed near the weld interface, where the thermal load is applied, and it decreases progressively away from the active weld zone. This behavior confirms that the heat transfer is strongest in the immediate vicinity of the welding region and becomes weaker in the remaining portions of the flange as the thermal energy spreads outward. The heat flux contours also help explain the development of the transient temperature field, since regions with high heat flux experience faster thermal response while regions farther away show slower heating. In the present study, the heat flux pattern is important because it provides a clear indication of the direction and intensity of thermal energy transfer during the welding cycle, which ultimately influences distortion, stress redistribution, and the overall thermo-mechanical behavior of the flange. A temperature probe was used at the weld region to support the interpretation of the thermal field and to correlate the heat flux distribution with the actual heating history.

2.5. Structural Boundary Conditions:

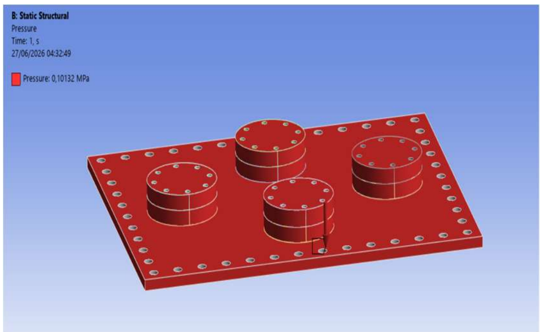


Figure 15. Atmospheric pressure boundary condition applied on the rectangular vacuum vessel flange model.

An atmospheric pressure boundary condition was applied on the exposed surface of the flange to represent the pressure difference between the external atmosphere and the vacuum environment inside the vessel. This loading generates an inward force on the flange and plays an important role in determining the overall structural response under service conditions. In the present study, atmospheric pressure was applied together with the imported thermal field so that the flange behavior could be evaluated under realistic thermo-mechanical loading.

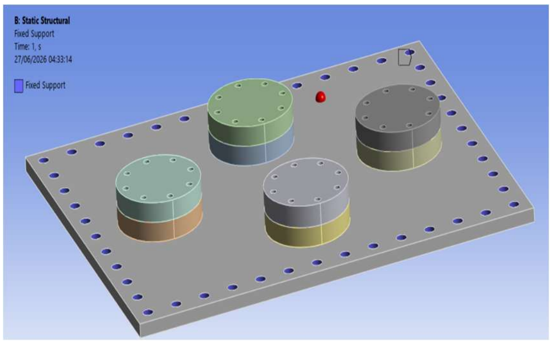


Figure 16. Fixed support boundary condition applied to the rectangular vacuum vessel flange model.

Fixed support was applied at the appropriate restrained region of the model to prevent rigid-body motion and to represent the structural constraint present in the actual assembly. This boundary condition ensures that the numerical model remains stable during analysis and that the deformation, stress, and strain results correspond to a physically meaningful response. The fixed support was therefore used as part of the static structural analysis to simulate the supported condition of the flange

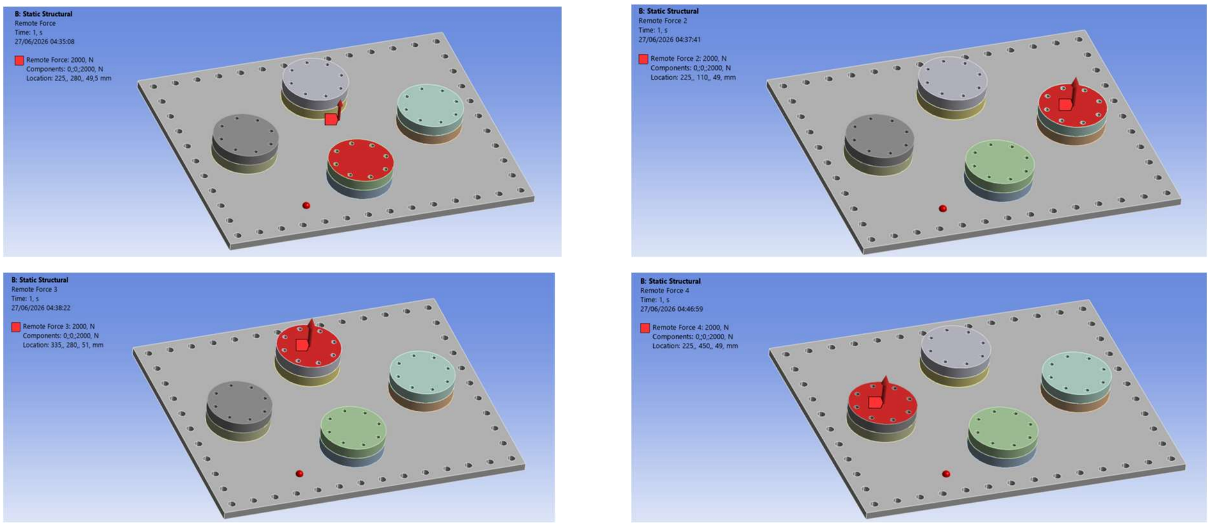


Figure 17. Remote force boundary condition applied to simulate equivalent axial clamping load (bolt pretension) on the flange model

A remote force was applied at the CF-port location to represent the equivalent axial clamping load produced by bolt pretension. In the actual flange assembly, this clamping action compresses the flange surfaces and contributes to the structural integrity of the joint. In the present model, the remote force was used to mimic this bolt-induced loading condition in a simplified but realistic manner, allowing the combined effect of clamping, thermal loading, and support restraint to be captured during static structural analysis.

- **Given:**

\- Bolt hole in CAD = Ø8.4 mm
(ISO 273 standard clearance for M8)
\- Bolt: M8, Grade 8.8
\- Nominal bolt diameter (d) = 8 mm
\- Thread pitch (p) = 1.25 mm (M8 coarse)
\- Proof Strength (Sp) = 600 MPa

- **Tensile Stress Area:**

$$At = (\pi/4) \times (d - 0.9382 \times p)^2$$
$$= (\pi/4) \times (8 - 0.9382 \times 1.25)^2$$
$$= (\pi/4) \times (8 - 1.173)^2$$
$$= (\pi/4) \times (6.827)^2$$
$$= 0.7854 \times 46.61$$
$$= 36.6 \text{ mm}^2$$

- **Pretension per bolt:**

$$Fi = 0.75 \times Sp \times At$$
$$= 0.75 \times 600 \times 36.6$$
$$= 16{,}470 \text{ N} \approx 15 \text{ kN per bolt}$$

- **Cross-check via Torque Method:**

T = 35 Nm (standard M8 Grade 8.8, dry)
K = 0.29 (nut factor, unlubricated)
d = 0.008 m (M8 nominal)

$$Fi = T / (K \times d)$$
$$= 35 / (0.29 \times 0.008)$$
$$= 35 / 0.00232$$
$$= 15{,}086 \text{ N} \approx 15 \text{ kN}$$

- **Total load per CF-63 port:**

$$Ftotal = 8 \text{ bolts} \times 15{,}000 \text{ N}$$
$$= 120{,}000 \text{ N} = 120 \text{ kN}$$

Theoretical bolt pretension = 120 kN per port. However, in this parametric study, the equivalent axial clamping load was applied as a remote force ranging from 1000 N to 10,000 N per port.

Justification:

The purpose of this parametric sweep is NOT to simulate the exact operational bolt load, but to study the STRUCTURAL SENSITIVITY of each thickness-sequence combination to varying clamping loads. Key principle: If the relative ranking of configurations remains consistent across the entire load range, the optimum selection is robust and load-independent.The results confirm that 16 mm + Sequence B shows the most balanced response across ALL load values in the range — validating the selection as load-independent optimum.

2.6. Static Structural Analysis:

Total Deformation:

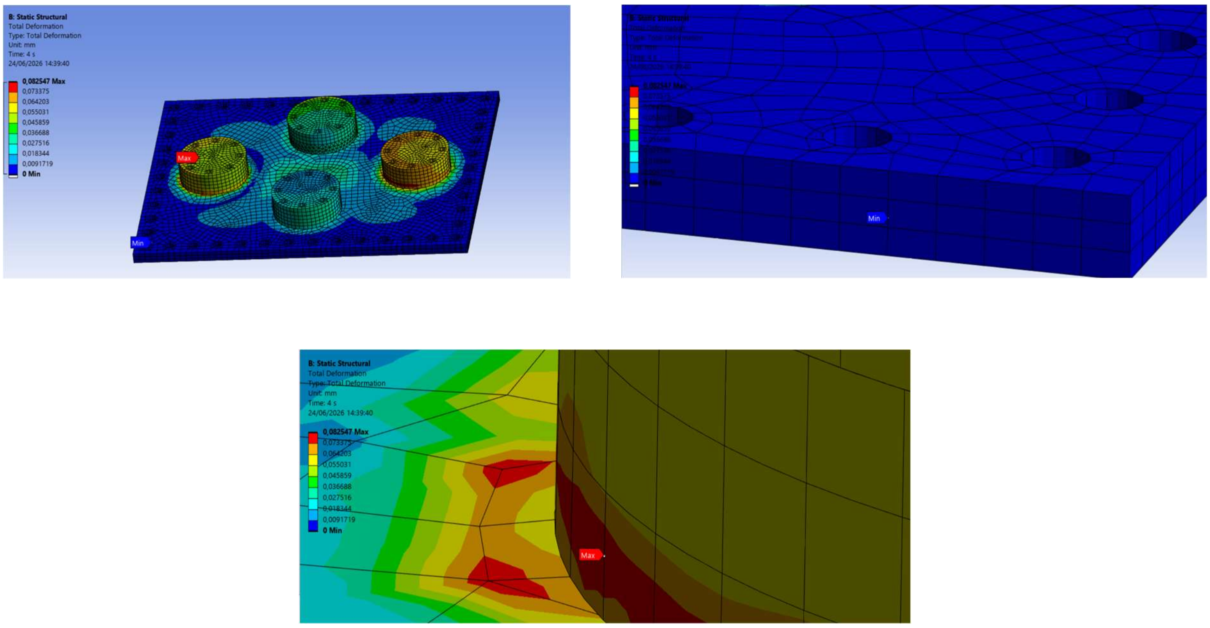


Figure 18. Total deformation distribution of the optimized 16 mm flange (Sequence B)

After completing the transient thermal analysis, the temperature field was imported into the static structural analysis to perform a coupled thermo-mechanical evaluation of the rectangular vacuum vessel flange. The structural model was subjected to atmospheric pressure, fixed support constraints, and an equivalent axial clamping load representing bolt pretension. Figure 13 shows the total deformation distribution of the optimized **16 mm flange welded using Sequence B (Symmetric/Opposite Side)** under the combined loading conditions. The deformation contour indicates that the maximum deformation occurs in the vicinity of the welded CF-port regions, where the combined effect of thermal expansion and mechanical loading is most significant. The remaining portion of the flange experiences comparatively lower deformation due to the support constraints and the inherent stiffness of the plate. The imported thermal field influences the structural response by introducing thermal expansion before the application of mechanical loads. The obtained deformation pattern demonstrates that the optimized flange configuration effectively controls distortion while maintaining structural stability. These results were further used for comparison with the remaining flange thicknesses during the optimization study.

Equivalent Von-Mises Stress:

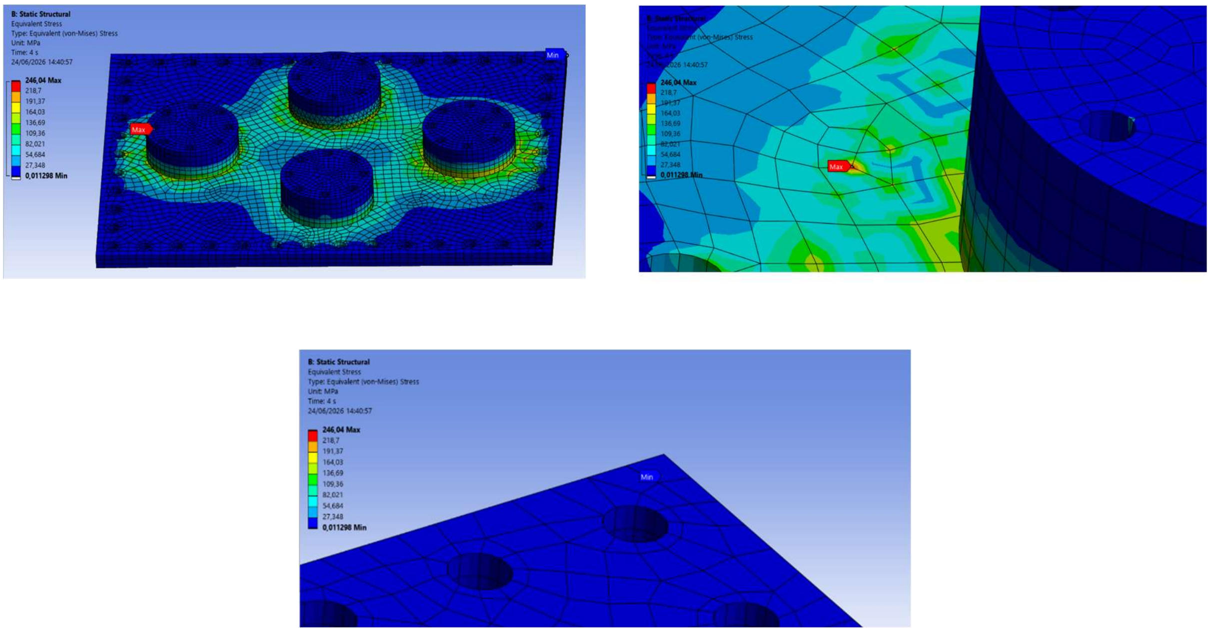


Figure 19. Equivalent von-Mises stress distribution of the optimized 16 mm flange (Sequence B)

The equivalent von-Mises stress distribution was evaluated to examine the stress developed in the flange after the application of thermo-mechanical loading. The structural analysis considered the combined influence of the imported thermal field, atmospheric pressure, fixed support, and the equivalent axial clamping load representing bolt pretension. Figure 14 presents the equivalent von-Mises stress distribution obtained for the optimized **16 mm flange using Sequence B**. The stress contour shows that higher stress values are mainly concentrated around the welded CF-port locations, where localized heating and mechanical constraints interact. Away from the weld region, the stress gradually decreases, resulting in a comparatively uniform stress distribution over the remaining portion of the flange. This behavior is expected because the welded region experiences the highest thermal gradient during the transient welding process. The obtained stress distribution provides a clear understanding of the structural response of the flange under actual operating conditions and serves as one of the important parameters for selecting the optimum flange configuration.

Equivalent Total Strain:

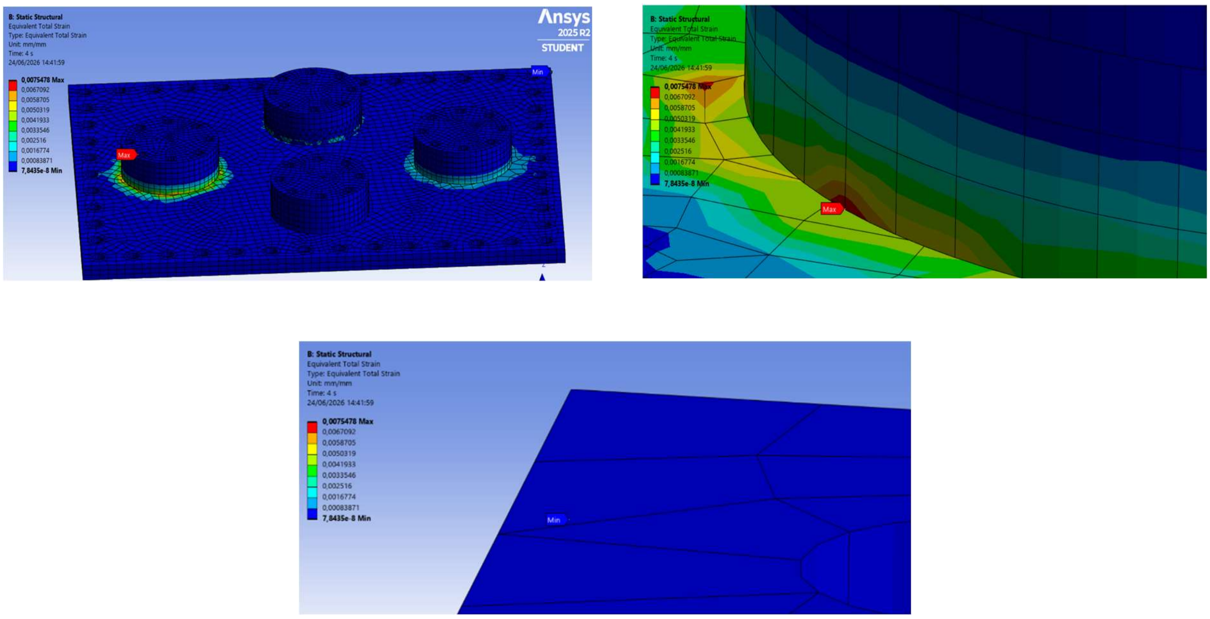


Figure 20. Equivalent total strain distribution of the optimized 16 mm flange (Sequence B)

The equivalent strain distribution was evaluated to study the deformation behaviour of the flange under the combined effect of thermal and structural loading. Figure 15 illustrates the equivalent strain distribution for the optimized **16 mm flange welded using Sequence B**. Similar to the stress distribution, the maximum strain is observed near the welded regions where the thermal expansion is highest during the welding process. As the distance from the weld interface increases, the strain gradually decreases because the thermal influence becomes less significant and the surrounding material provides structural restraint. The overall strain distribution indicates that deformation is primarily localized near the weld region, while the remaining portion of the flange undergoes relatively small strain. Together with the deformation and von-Mises stress results, the equivalent strain distribution was used to compare different flange thicknesses and welding sequences and to identify the optimum design configuration.

2.7. Comparative Structural Analysis:

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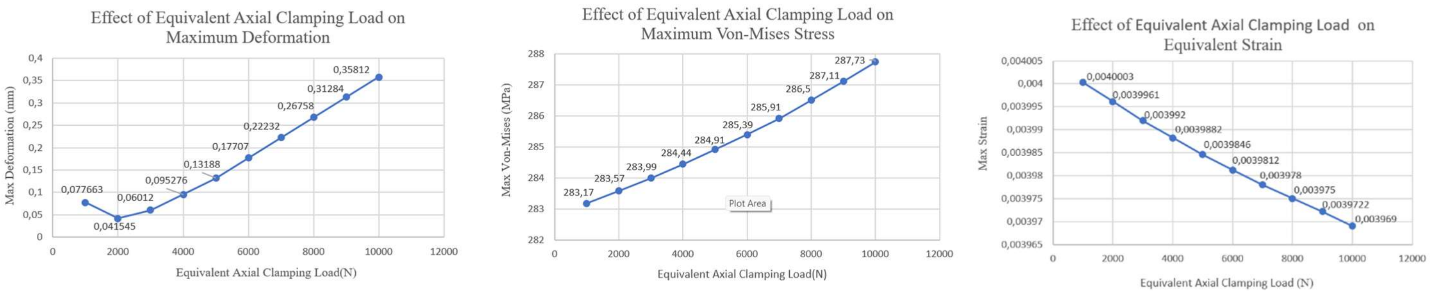

Figure 21.

Variation of maximum deformation, equivalent von-Mises stress, and equivalent strain with equivalent axial clamping load for the 13 mm rectangular vacuum vessel flange using Sequence C (Alternate Cross-Pattern).

The structural response of the **13 mm** flange was evaluated using **Sequence C (Alternate Cross-Pattern)** under different equivalent axial clamping loads. As the clamping load increased, the total deformation increased gradually due to the additional compressive loading introduced by bolt pretension. A continuous increase in equivalent von-Mises stress was also observed with increasing clamping load, indicating greater stress concentration around the flange and welded regions. In contrast, the equivalent strain decreased progressively with increasing clamping load, reflecting improved structural restraint under higher bolt preload. Although the **13 mm** flange exhibited the **minimum deformation** among all the investigated thicknesses, the comparatively higher stress values indicated that this configuration was not the optimum choice when all structural parameters were considered simultaneously

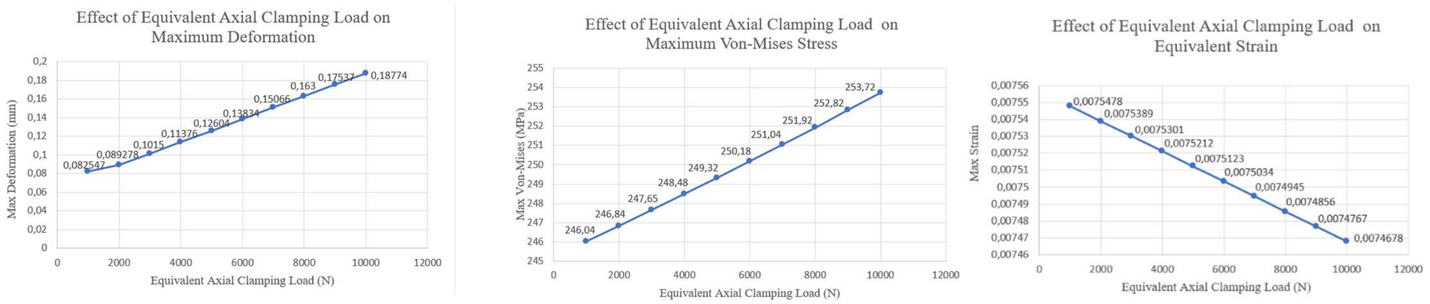

Figure 22.

Variation of maximum deformation, equivalent von-Mises stress, and equivalent strain with equivalent axial clamping load for the 16 mm rectangular vacuum vessel flange using Sequence B (Symmetric/Opposite Side).

The **16 mm** flange was analysed using **Sequence B (Symmetric/Opposite Side)** for different equivalent axial clamping loads. The deformation increased gradually with increasing clamping load, following the expected structural response under increasing bolt pretension. Similarly, the equivalent von-Mises stress showed a moderate increase as the clamping load increased, while the equivalent strain reduced steadily due to improved restraint of the structure. Compared with the other flange thicknesses, this configuration provided the most balanced combination of deformation, stress, and strain. The results indicate that the **16 mm flange with Sequence B** effectively distributes thermo-mechanical loads while maintaining structural integrity, making it the optimum configuration selected in this study.

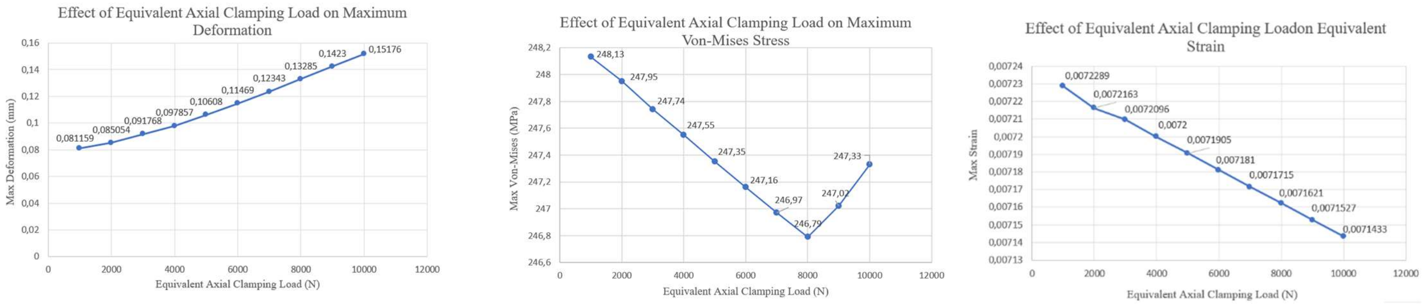

Figure 23.

Variation of maximum deformation, equivalent von-Mises stress, and equivalent strain with equivalent axial clamping load for the 19 mm rectangular vacuum vessel flange using Sequence C (Alternate Cross-Pattern).

The structural performance of the **19 mm** flange was investigated using **Sequence C (Alternate Cross-Pattern)** under varying equivalent axial clamping loads. The deformation increased gradually as the clamping load increased, similar to the behaviour observed for the other flange thicknesses. The equivalent von-Mises stress initially decreased with increasing clamping load before showing a slight rise at the highest loading level, indicating a redistribution of stresses within the flange. The equivalent strain continuously decreased with increasing clamping load, suggesting improved stiffness under higher preload conditions. Although the **19 mm** flange exhibited lower stress than the 13 mm configuration, its overall structural response was not superior to the optimized **16 mm** flange when deformation, stress, and strain were evaluated together.

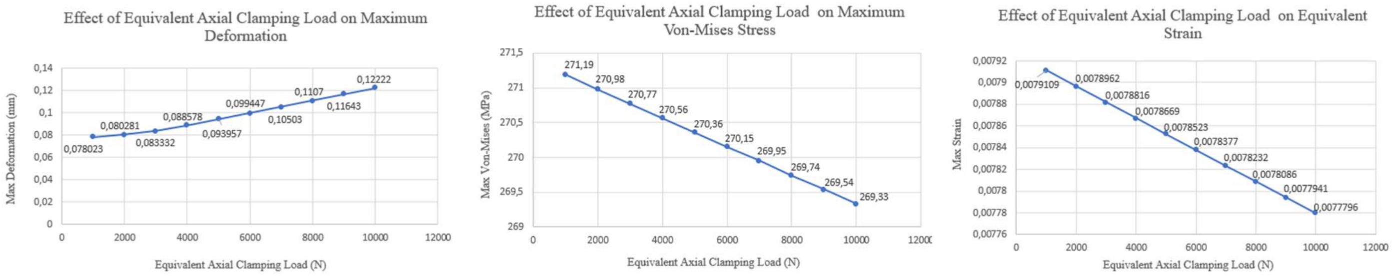

Figure 24.

Variation of maximum deformation, equivalent von-Mises stress, and equivalent strain with equivalent axial clamping load for the 22 mm rectangular vacuum vessel flange using Sequence C (Alternate Cross-Pattern).

The **22 mm** flange was analysed using **Sequence C (Alternate Cross-Pattern)** for different equivalent axial clamping loads. The deformation increased progressively with increasing clamping load, whereas the equivalent von-Mises stress decreased gradually throughout the loading range. A similar decreasing trend was observed for the equivalent strain, indicating enhanced structural rigidity with increasing bolt preload. Although the larger plate thickness improved stiffness, the overall structural performance did not provide a significant advantage over the optimized **16 mm** flange. The increased thickness resulted in additional material usage without producing a better combined response in terms of deformation, stress, and strain. Therefore, the **22 mm** configuration was not selected as the optimum design.

3.Final Optimization Results:

The final optimization results were obtained by comparing the optimum structural response corresponding to each plate thickness after selecting the most suitable welding sequence and equivalent axial clamping load. This comparison represents the final design selection stage of the study, where the best-performing configuration for each thickness was evaluated using the three structural performance parameters.

The deformation comparison shows that the **13 mm flange** achieved the lowest deformation; however, it exhibited comparatively higher equivalent von-Mises stress. Increasing the plate thickness to **16 mm** resulted in only a small increase in deformation while producing a significant reduction in equivalent von-Mises stress. This indicates a more favourable balance between structural stiffness and thermo-mechanical loading.

For the **19 mm** and **22 mm** flanges, no significant improvement in the combined structural response was observed. Although the structural behaviour changed with increasing plate thickness, neither configuration simultaneously reduced deformation, stress, and strain compared with the optimized 16 mm flange.

The strain comparison further confirms that selecting the optimum flange cannot be based on deformation or stress alone. A comprehensive evaluation of deformation, equivalent von-Mises stress, and equivalent strain is essential to identify the most suitable design.

Based on the complete optimization study, the **16 mm flange welded using Sequence B (Symmetric/Opposite Side)** and subjected to an equivalent axial clamping load of **1000 N** was identified as the optimum configuration. This design provided the best overall balance between deformation control, stress reduction, and strain distribution, making it the recommended flange configuration for the present study.

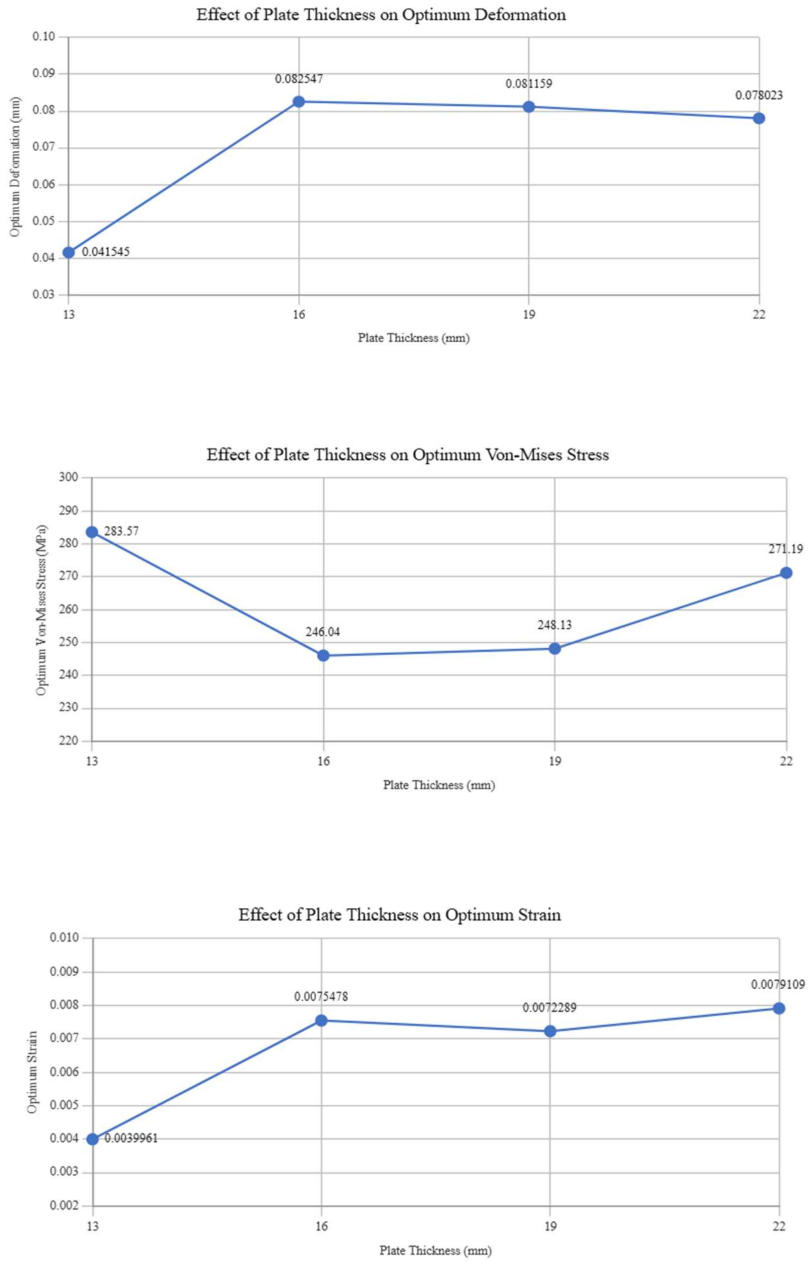

Figure 25. Comparison of optimum deformation, equivalent von-Mises stress, and equivalent strain for different plate thicknesses after optimization of welding sequence and equivalent axial clamping load.

4.Conclusion:

The present work successfully investigated the thermo-mechanical behaviour of a rectangular vacuum vessel flange used in spherical tokamak applications through finite element analysis. Parametric CAD models having plate thicknesses of **13 mm, 16 mm, 19 mm, and 22 mm** were developed in Autodesk Fusion 360 and analysed in ANSYS Mechanical using a coupled transient thermal and static structural workflow.

A moving welding heat source was applied during transient thermal analysis together with convection and radiation boundary conditions to simulate the welding process. The resulting temperature field was subsequently imported into the static structural analysis, where atmospheric pressure, fixed support conditions, and an equivalent axial clamping load representing bolt pretension were applied to reproduce realistic operating conditions.

A detailed comparative study was carried out by investigating three welding sequences and different equivalent axial clamping loads for each plate thickness. The structural response of each configuration was evaluated in terms of **total deformation, equivalent von-Mises stress, and equivalent strain**. The results demonstrate that plate thickness, welding sequence, and bolt pretension collectively influence the structural behaviour of the rectangular vacuum vessel flange.

Among all the configurations analysed, the **16 mm flange welded using Sequence B (Symmetric/Opposite Side)** with an equivalent axial clamping load of **1000 N** provided the most balanced structural response. Although the **13 mm flange** produced the minimum deformation, it also generated comparatively higher equivalent von-Mises stress. Similarly, the **19 mm** and **22 mm** flanges did not provide a better combined structural response than the optimized 16 mm configuration. Therefore, the final design selection was based on the simultaneous consideration of deformation, equivalent von-Mises stress, and equivalent strain rather than on any individual parameter.

The methodology developed in this project demonstrates that finite element analysis is an effective engineering tool for evaluating welded flange designs before manufacturing. The workflow established in this study can be used as a systematic approach for future thermo-mechanical optimization of vacuum vessel flanges and similar welded components employed in fusion engineering applications.

Safety Factor Verification (16mm, Seq B):

SF = UTS / Max Von Mises Stress
= 485 MPa / 246 MPa
= 1.97 ≈ 2.0

This confirms that while local yielding occurs at weld toe (246 MPa > 170 MPa yield), the global plate remains within safe structural limits with adequate margin against ultimate failure.

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